

PROGRAMMING PRINCIPLES

PROJECT MANAGEMENT, ASSIGNMENT IDEA

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NO MID-TERM = DO EXERCISES

ATTENDANCE: 10%

MID-TERM: 20%

ASSIGNMENT: 30%

FINAL EXAM: 40%

PROJECT ASSIGNMENT GUIDELINES

- **TEAM COMPOSITION**
 - FORM A GROUP OF **4–5 MEMBERS**.
- **PROJECT TOPIC & PURPOSE**
 - SELECT A **SPECIFIC TOPIC** FOR THE ASSIGNMENT.
 - CLEARLY **EXPLAIN THE PURPOSE** AND RELEVANCE OF THE PROJECT.
- **PLATFORM & TECHNOLOGY**
 - DEVELOP A **WEB APPLICATION** USING **JAVA**, **USE AI TO CREATE OR ASSIST**
 - IMPLEMENT **1–2 FUNCTIONAL MODULES** WITH **FULL CRUD OPERATIONS** (CREATE, READ, UPDATE, DELETE).
 - ENSURE A **SIMPLE, CLEAN, AND USER-FRIENDLY UI**.
 - CREATE A **WELL-STRUCTURED DATABASE DESIGN**.
 - USE **JAVA CLASSES** WITH PROPER **OBJECT-ORIENTED DESIGN** PRINCIPLES.
- **PROJECT EXPLANATION & DOCUMENTATION**
 - **IDENTIFY AND DESCRIBE THE ISSUE/PROBLEM** YOUR PROJECT ADDRESSES.
 - PROVIDE A **CLEAR SOLUTION APPROACH**.
 - **DETAIL HOW THE SOLUTION IS IMPLEMENTED** (TECHNICAL WORKFLOW, TOOLS, FRAMEWORKS).

SIMPLE PROJECT MANAGEMENT FOR JAVA + AI PROJECTS

OVERVIEW:

- MANAGE SMALL PROJECTS EFFECTIVELY
- WORK IN TEAMS
- TRACK WORKFLOW AND TIMELINES
- APPLY CONCEPTS IN CAMBODIA'S CONTEXT WITH JAVA & AI



PROJECT



BUSINESS MEETING



VISION



CASH FLOW



IMPLEMENTATION



SCHEDULE



COLLABORATION



BUDGET



MANAGER



PROJECT
DOCUMENTATION



STRATEGY



CLIENT



TIME MANAGEMENT



TEAM



DEVELOPMENT



ANALYTICS



WORKFLOW



OBJECTIVES



DECISION MAKING



PROJECT
MANAGEMENT

SIMPLE PROJECT MANAGEMENT OVERVIEW

**DEFINITION: COORDINATING TASKS, PEOPLE, AND RESOURCES TO
ACHIEVE GOALS ON TIME**

CORE ELEMENTS:

1. SCOPE – WHAT TO BUILD
2. RESOURCES – PEOPLE, TOOLS, AND BUDGET
3. TIME – DEADLINES AND MILESTONES
4. QUALITY – MEETING USER REQUIREMENTS



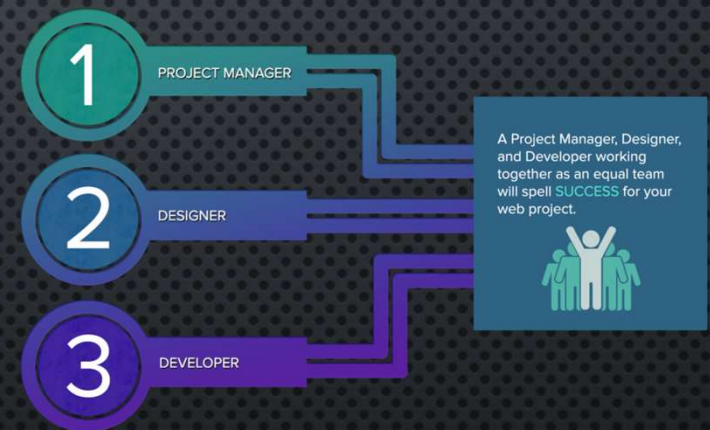
TEAM MANAGEMENT BASICS

ROLES IN A SMALL TEAM:

- PROJECT MANAGER: COORDINATES TASKS & TIMELINES
- DEVELOPER(S): IMPLEMENT FEATURES
- UI/UX DESIGNER: DESIGNS USER INTERFACE
- TESTER: VALIDATES FEATURES
- DOCUMENTATION LEAD: KEEPS RECORDS

TIPS:

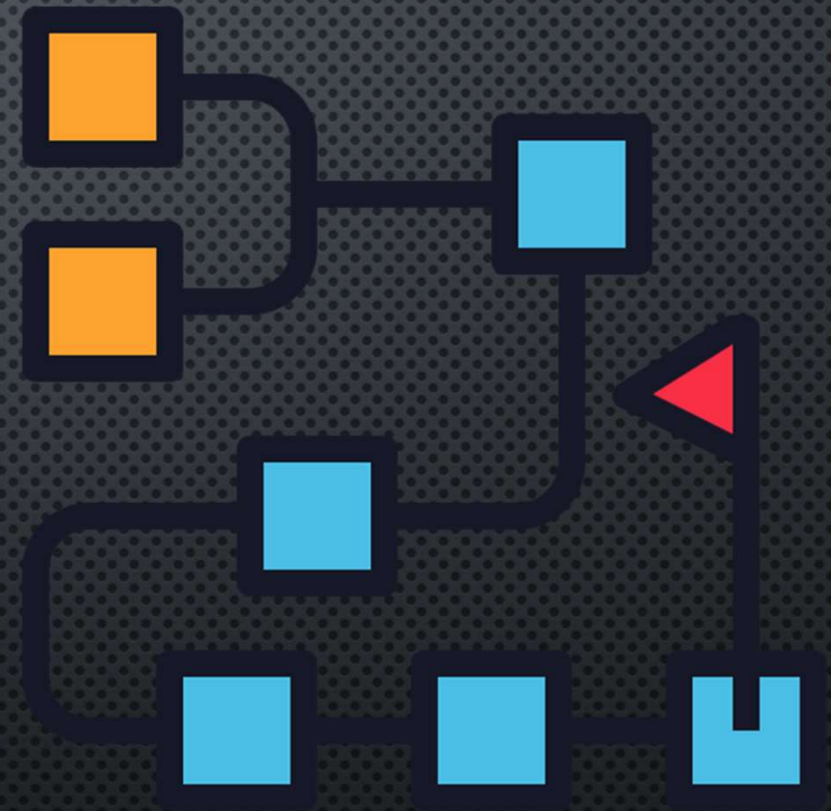
- CLEAR ROLE ASSIGNMENTS
- WEEKLY CHECK-INS
- SHARED TASK BOARD (TRELLO, GITLAB, GITHUB PROJECTS)



WORKFLOW MANAGEMENT

PHASES:

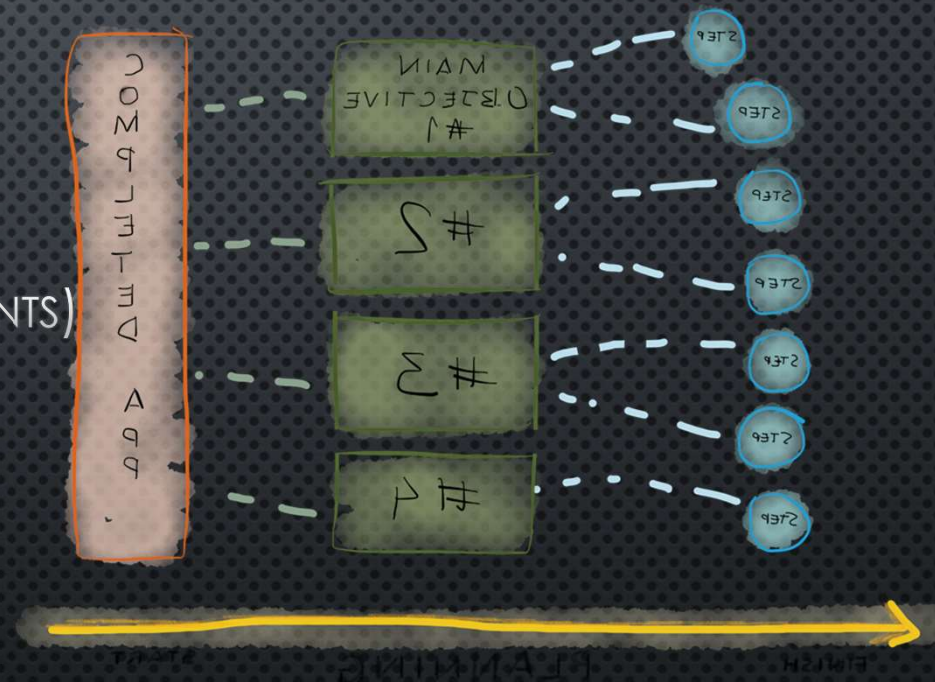
- 1. REQUIREMENT GATHERING
- 2. PLANNING & DESIGN (UML, DB SCHEMA)
- 3. IMPLEMENTATION (JAVA + AI INTEGRATION)
- 4. TESTING (UNIT & INTEGRATION TESTS)
- 5. DEPLOYMENT & REVIEW
- TOOLS: GIT, TRELLO/JIRA, GOOGLE DOCS



TIMELINE MANAGEMENT

EXAMPLE TIMELINE (8 WEEKS):

- WEEK 1–2: REQUIREMENTS & DESIGN
- WEEK 3–5: DEVELOPMENT (SPLIT INTO SPRINTS)
- WEEK 6: INTERNAL TESTING
- WEEK 7: USER TESTING & BUG FIXES
- WEEK 8: FINAL DELIVERY & PRESENTATION
- TIP: USE GANTT CHART FOR CLARITY



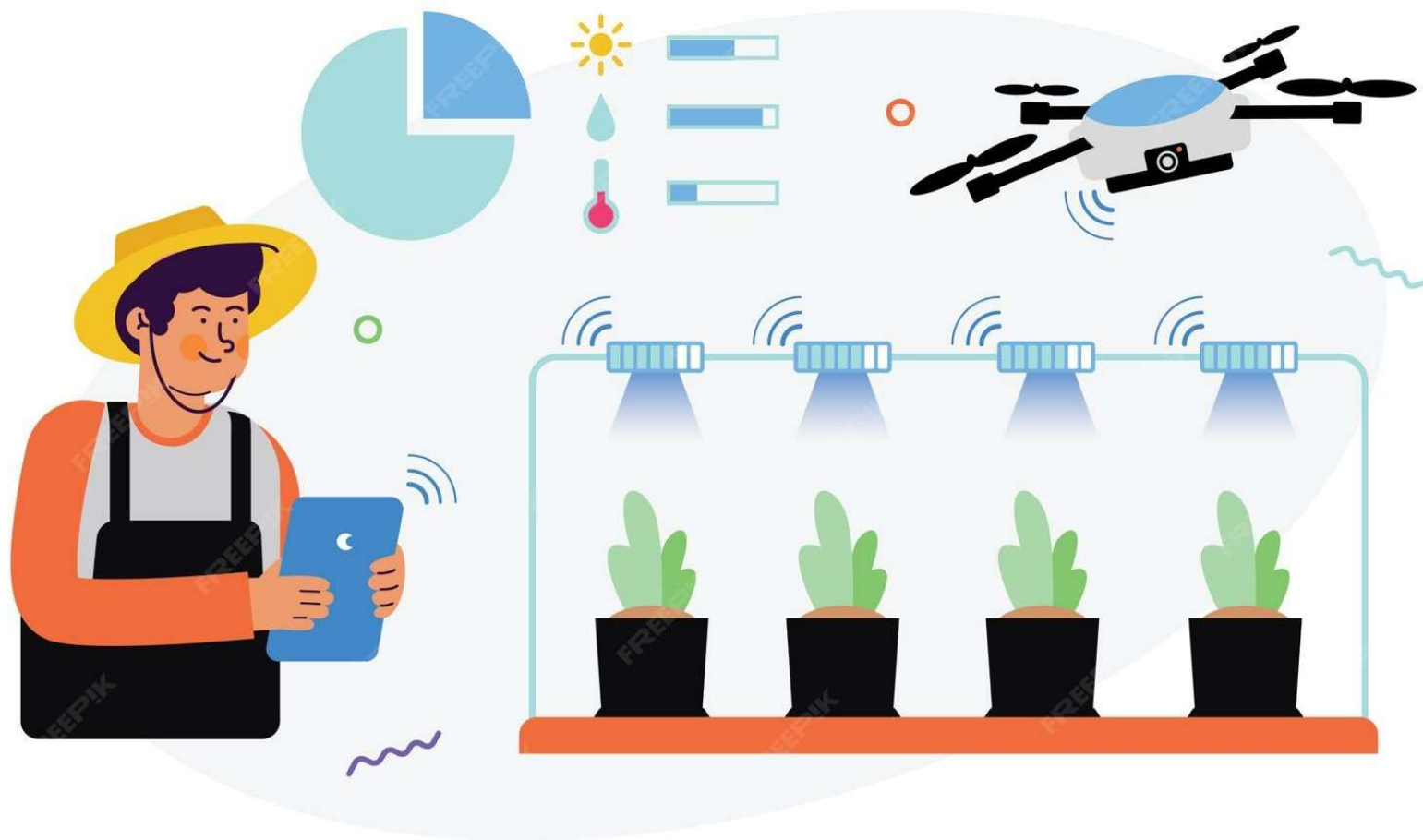
CAMBODIA-CONTEXT PROJECT IDEAS (JAVA + AI)

1. SMART CROP DISEASE DETECTOR – FARMERS UPLOAD PLANT PHOTOS, AI DETECTS DISEASE
2. CAMBODIA TRAFFIC VIOLATION TRACKER – AI FROM CCTV IMAGES
3. KHMER DOCUMENT SUMMARIZER – AI SUMMARIZES KHMER TEXT
4. LOCAL MARKET PRICE TRACKER – AI PREDICTS AGRI PRODUCT TRENDS
5. SMART CLASS ATTENDANCE – AI FACIAL RECOGNITION FOR SCHOOLS

AI in Agriculture

EPS File

Vector Illustration



EXAMPLE: SMART CROP DISEASE DETECTOR WORKFLOW

1. FARMER UPLOADS PHOTO VIA WEB INTERFACE
2. IMAGE SENT TO AI API FOR CLASSIFICATION
3. RESULT + SUGGESTED SOLUTION STORED IN DB
4. FARMER CAN VIEW HISTORY OF DIAGNOSES

SUMMARY

- START SIMPLE, PLAN CLEARLY, SPLIT RESPONSIBILITIES
- USE VISUAL TOOLS FOR TRACKING PROGRESS
- KEEP TIMELINE REALISTIC BUT DISCIPLINED
- CHOOSE PROJECTS WITH LOCAL IMPACT FOR MOTIVATION